Chapter 11. Analyzing Repeated Categorical Response Data

Marginal models, GEE, Markov chains

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Repeated responses arise over time, under several conditions (crossover trials) or within *clusters* (teeth, litters, families), and are dependent within a cluster.

11.1 Comparing marginal distributions: multiple responses

Motivation. The T margins usually matter more than the joint dependence.

11.1.1 Binary marginal models and marginal homogeneity

For binary $(Y_1, \ldots, Y_T)$ the marginal logit model

$$\operatorname{logit} P(Y_t = 1) = \alpha + \beta_t, \qquad t = 1, \ldots, T, \tag{11.1}$$

($\beta_T = 0$) extends Section 10.2.1. Let $\pi_{\mathbf{i}} = P(Y_1 = i_1, \ldots, Y_T = i_T)$ in the 2^T table. (11.1) is saturated; **marginal homogeneity** is $\beta_1 = \cdots = \beta_T$, awkward to fit by ML since the likelihood involves the 2^T joint probabilities. With $p_{\mathbf{i}} = n_{\mathbf{i}}/n$ and $\hat{\pi}^{MH}$ the constrained ML fit,

$$-2\big[L(\hat{\pi}^{MH}) - L(\mathbf{p})\big] = 2\sum_{\mathbf{i}} n_{\mathbf{i}} \log\left(p_{\mathbf{i}}/\hat{\pi}^{MH}_{\mathbf{i}}\right) \tag{11.2}$$

is asymptotically χ^2_{T-1} under H_0.

11.1.2 Crossover drug comparison example

Example 11.1 (Crossover study of three drugs)

46 subjects, drugs A, B, C, each response favorable or not. Proportions favorable $(0.61, 0.61, 0.35)$; (11.2) $= 5.95$ (df $= 2$, $P = 0.05$). Bonferroni interval ($z = 2.39$), A vs C: $0.261 \pm 2.39(0.108) = (0.002, 0.520)$ (Theorem 10.1); same for B vs C.

11.1.3 Modeling margins of a multicategory response

For I categories, the saturated model $\log[P(Y_t = j)/P(Y_t = I)] = \beta_{tj}$ gives a marginal homogeneity test ($\beta_{1j} = \cdots = \beta_{Tj}$) of the form (11.2) with df $= (T-1)(I-1)$. For ordinal responses,

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta_t, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1, \tag{11.3}$$

($\beta_T = 0$) gives a test of $\beta_1 = \cdots = \beta_T$ with df $= T - 1$.

11.1.4 Wald and generalized CMH score tests of marginal homogeneity

Let $p_j(t)$ be the sample proportion in category j at occasion t, $d_j(t) = p_j(t) - \sum_s p_j(s)/T$, and $\mathbf{d}$ the vector of $d_j(t)$, $t \le T-1$, $j \le I-1$. Bhapkar's statistic

$$W = n\,\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} \tag{11.4}$$

is asymptotically $\chi^2_{(I-1)(T-1)}$, generalizing Proposition 10.14.

Generalized CMH statistics treat each subject as a $T \times I$ stratum. Conditional independence given subject implies marginal homogeneity; for $I = T = 2$ this is McNemar (Theorem 10.5), for $I = 2$, $T > 2$ Cochran's Q.

Theorem 11.2 (Cochran's Q as a generalized CMH statistic)

For binary responses, let R_k be the number of successes of subject k, C_t the number of successes at occasion t and $N = \sum_t C_t = \sum_k R_k$. The generalized CMH statistic (general association) for the $T \times 2 \times n$ subject-stratified table is

$$Q = \frac{T(T-1)\sum_{t=1}^{T}(C_t - N/T)^2}{TN - \sum_k R_k^2},$$

which is asymptotically χ^2_{T-1} under marginal homogeneity. For $T = 2$, $Q = (n_{21}-n_{12})^2/(n_{21}+n_{12})$.

Proof. [Supplement]

Step 1. Under conditional independence with fixed stratum margins, subject k's R_k successes are allocated to the T occasions completely at random, independently over subjects; the CMH statistic is the quadratic form in $\mathbf{C} = (C_1, \ldots, C_T)'$ about its null mean with a generalized inverse of its null covariance.

Step 2. With $c_k = (R_k/T)(1 - R_k/T)$, sampling without replacement gives $E(Y_{kt}) = R_k/T$, $\mathrm{Var}(Y_{kt}) = c_k$, $\mathrm{Cov}(Y_{kt}, Y_{ks}) = -c_k/(T-1)$, i.e. $\mathrm{Cov}(\mathbf{Y}_k) = \frac{T}{T-1} c_k \mathbf{P}$ with $\mathbf{P} = \mathbf{I} - \frac{1}{T}\mathbf{J}$.

Step 3. Summing over subjects, $E(\mathbf{C}) = (N/T)\mathbf{1}$ and $\mathrm{Cov}(\mathbf{C}) = \kappa \mathbf{P}$ with

$$\kappa = \frac{T}{T-1}\sum_k c_k = \frac{TN - \sum_k R_k^2}{T(T-1)}.$$

Step 4. $\mathbf{P}$ is the projection onto $\{\mathbf{1}'\mathbf{v} = 0\}$, its own generalized inverse, and contains $\mathbf{C} - E\mathbf{C}$ in its range, so the quadratic form is $\kappa^{-1}\sum_t (C_t - N/T)^2 = Q$. Since $\mathrm{rank}\,\mathbf{P} = T - 1$, the CLT gives the χ^2_{T-1} limit.

Step 5. For $T = 2$, concordant subjects contribute 0 to $2N - \sum_k R_k^2$, which equals $n_{12} + n_{21}$, and $\sum_t (C_t - N/2)^2 = (n_{12} - n_{21})^2/2$; hence $Q = (n_{12} - n_{21})^2/(n_{12} + n_{21})$. ■

11.2 Marginal modeling: maximum likelihood approach

Motivation. ML for marginal logit models maximizes the multinomial likelihood under constraints on the margins.

11.2.1 Longitudinal mental depression example

Example 11.3 (Mental depression)

Mild or severe depression ($s = 0, 1$), standard or new drug ($d = 0, 1$), normal or abnormal at weeks 1, 2, 4 ($t = 0, 1, 2$, $\log_2$ week); $n = 340$. The proportion normal rises faster for the new drug (0.18 → 0.83 severe/new, 0.21 → 0.46 severe/standard). Main effects: $G^2 = 34.6$ (df = 8); adding drug × time,

$$\mathrm{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t + \beta_4 dt,$$

gives $G^2 = 4.2$ (df = 7), with $\hat\beta_3 = 0.48$ (SE 0.12), $\hat\beta_4 = 1.01$ (SE 0.18), $\hat\beta_1 = -1.29$ (SE 0.14) and $\hat\beta_2 = -0.06$ (SE 0.22).

11.2.2 Modeling a repeated multinomial response

For I categories, model the $I-1$ marginal logits at each t as $\text{logit}_j(t) = \alpha_j + \boldsymbol{\beta}_j'\mathbf{x}_t$ ($\boldsymbol{\beta}_j = \boldsymbol{\beta}$ under proportional odds), checking whether effects change over t.

11.2.3 Insomnia example

Example 11.4 (Insomnia clinical trial)

Time to fall asleep (4 ordered categories) before ($t=0$) and after ($t=1$) two weeks of active treatment ($x=1$) or placebo ($x=0$). The proportional odds model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta_1 t + \beta_2 x + \beta_3 tx \tag{11.5}$$

has $G^2 = 8.0$ (df $= 6$), $\hat\beta_1 = 1.074$ (SE 0.162), $\hat\beta_2 = 0.046$ (SE 0.236), $\hat\beta_3 = 0.662$ (SE 0.244): active-vs-placebo cumulative odds ratio 1.04 initially, $e^{0.708} = 2.03$ at follow-up.

11.2.4 Comparisons that control for initial response

If initial margins differ between groups, marginal changes may differ even with identical conditional follow-up distributions; then compare follow-up given y_1,

$$\text{logit}\, P(Y_2 \le j) = \alpha_j + \beta_1 x + \beta_2 y_1, \tag{11.6}$$

an ordinal analogue of ANCOVA and a simple transitional model (Section 11.5).

11.3 Marginal modeling: generalized estimating equations (GEE) approach

Motivation. GEE needs only a marginal mean, a variance function and a working correlation, and still estimates $\boldsymbol{\beta}$ consistently.

11.3.1 Generalized estimating equation methodology: basic ideas

For $(Y_1, \ldots, Y_T)$ (T may vary by subject), specify a link for $\mu_t = E(Y_t)$, a variance function $v(\mu_t)$ and a *working correlation*; the estimates solve the **generalized estimating equations** (Liang and Zeger 1986).

Definition (Working correlation structures)

For cluster i with T_i observations, a working correlation matrix $\mathbf{R}(\boldsymbol{\alpha})$ specifies $\text{Corr}(Y_{it}, Y_{is})$:
(i) *independence*: $\mathbf{R} = \mathbf{I}$;
(ii) *exchangeable*: $\text{Corr}(Y_{it}, Y_{is}) = \alpha$ for all $t \neq s$;
(iii) *autoregressive* AR(1): $\text{Corr}(Y_{it}, Y_{is}) = \alpha^{|t-s|}$;
(iv) *unstructured*: a separate α_{ts} for each pair.

The key properties, proved in Section 11.4:

(a) If the mean model is correct, GEE estimators are consistent *even if the working correlation is wrong.*

(b) Naive standard errors are then wrong; *robust* (sandwich) standard errors use the empirical dependence.

Even independence gives good estimates with corrected SEs; a good working correlation improves efficiency. Compare structures for sensitivity; exchangeable is a sensible default, unstructured costs efficiency for large T.

Limitations of GEE

There is no likelihood, so inference uses Wald statistics with the sandwich covariance. With few clusters sandwich SEs are too small and variable; quasi-score tests (Boos 1992; Rotnitzky and Jewell 1990) may be better.

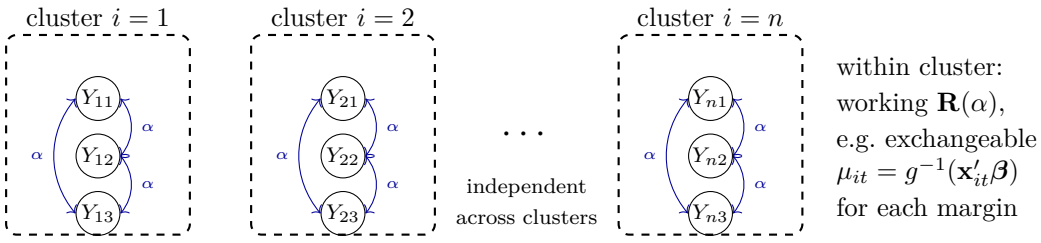


Figure 11.1. n independent clusters with dependent observations; GEE asymptotics are in n.

11.3.2 Longitudinal mental depression example

Example 11.5 (Depression: GEE fit)

Example 11.3: GEE is close to ML for every working correlation; exchangeable gives $\hat{\beta}_3 = 0.48$ (SE 0.12), $\hat{\beta}_4 = 1.02$ (SE 0.19). Independence reproduces ordinary logistic regression on 3×340 observations (intercept -0.028, severe -1.314, new drug -0.060, time 0.482, drug $\times$ time 1.017); naive SEs $(0.164, 0.146, 0.222, 0.115, 0.189)$ and robust $(0.174, 0.146, 0.229, 0.120, 0.188)$ nearly agree as $\hat{\alpha}_{\text{exch}} = -0.003$.

11.3.3 GEE approach for multinomial responses: insomnia example

For multinomial responses (Lipsitz, Kim and Zhao 1994), choose a working correlation for each pair of categories across occasions, with the $(I-1) \times (I-1)$ multinomial covariance at each occasion.

Example 11.6 (Insomnia: GEE fit)

For (11.5) with independence working correlation, $\hat{\beta}_1 = 1.038$ (SE 0.168), $\hat{\beta}_2 = 0.034$ (SE 0.238), $\hat{\beta}_3 = 0.708$ (SE 0.244), similar to ML.

11.4 Quasi-likelihood and its GEE multivariate extension: details

Motivation. Consistency under a wrong correlation and the sandwich both follow from one fact about mean-zero estimating functions (Lemma 11.8).

For subject i let y_i have mean μ_i, variance function $v(\mu_i)$, and $\eta_i = g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$. The **quasi-likelihood** (QL) estimates solve

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big(\frac{\partial \mu_i}{\partial \boldsymbol{\beta}}\Big) \frac{y_i - \mu_i}{v(\mu_i)} = \mathbf{0}, \qquad \mu_i = g^{-1}(\mathbf{x}_i'\boldsymbol{\beta}). \tag{11.7}$$

Proposition 11.7 (Motivations for the quasi-score equations)

(a) If y_i has a natural exponential family distribution with variance function v, (11.7) are the GLM likelihood equations.
(b) With $v(\mu_i)$ replaced by known constants v_i, (11.7) are the normal equations for minimizing $\sum_i (y_i - \mu_i)^2 / v_i$.
(c) Since $\partial \mu_i / \partial \beta_j = (\partial \mu_i / \partial \eta_i) x_{ij}$, (11.7) depend only on the link and the variance function.

Proof. [Supplement] (a) For $f(y;\theta) = \exp\{y\theta - b(\theta) + c(y)\}$, $\mu = b'(\theta)$ and $\partial\mu/\partial\theta = b''(\theta) = v(\mu)$, so

$$\frac{\partial [y_i \theta_i - b(\theta_i)]}{\partial \beta_j} = (y_i - \mu_i) \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \beta_j} = \frac{y_i - \mu_i}{v(\mu_i)} \frac{\partial \mu_i}{\partial \beta_j}.$$

(b) $\partial / \partial \beta_j$ of $\sum_i (y_i - \mu_i)^2 / v_i$ is $-2 \sum_i (y_i - \mu_i) v_i^{-1} \partial \mu_i / \partial \beta_j$.

(c) Chain rule. ■

11.4.1 Properties of quasi-likelihood estimators

Definition (Unbiased estimating function)

A function $\mathbf{h}(\mathbf{y}; \boldsymbol{\beta})$ is an **unbiased estimating function** if $E_{\boldsymbol{\beta}}[\mathbf{h}(\mathbf{Y}; \boldsymbol{\beta})] = \mathbf{0}$ for all $\boldsymbol{\beta}$; the equations $\mathbf{h}(\mathbf{y}; \boldsymbol{\beta}) = \mathbf{0}$ that define $\hat{\boldsymbol{\beta}}$ are **estimating equations**.

The quasi-score is unbiased whenever $E(y_i) = \mu_i(\boldsymbol{\beta})$, whatever the true variance.

Lemma 11.8 (Asymptotics of estimating equations)

Let clusters $i = 1, \ldots, n$ be independent, $\boldsymbol{\beta} \in \Theta \subseteq \mathbb{R}^p$ open, with true value $\boldsymbol{\beta}_0$, and let $\boldsymbol{\gamma}$ be a nuisance parameter estimated by $\hat{\boldsymbol{\gamma}} \xrightarrow{p} \boldsymbol{\gamma}^*$. Let $\mathbf{U}_n(\boldsymbol{\beta}; \boldsymbol{\gamma}) = \sum_{i=1}^n \boldsymbol{\psi}_i(\boldsymbol{\beta}; \boldsymbol{\gamma})$ and $\mathbf{H}_i = -\partial \boldsymbol{\psi}_i / \partial \boldsymbol{\beta}'$. Assume:
(E1) $E\, \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) = \mathbf{0}$ for all i;
(E2) $n^{-1} \sum_i \operatorname{Cov}\left(\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*)\right) \to \mathbf{B}$, positive definite, and $\sup_i E \|\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*)\|^{2+\delta} < \infty$ for some $\delta > 0$;
(E3) $\boldsymbol{\psi}_i$ is continuously differentiable in $\boldsymbol{\beta}$, and on neighbourhoods $\mathcal{N}$ of $\boldsymbol{\beta}_0$ and $\mathcal{M}$ of $\boldsymbol{\gamma}^*$, $\sup_{\mathcal{N} \times \mathcal{M}} \| n^{-1} \sum_i \mathbf{H}_i(\boldsymbol{\beta}; \boldsymbol{\gamma}) - \mathbf{A}(\boldsymbol{\beta}, \boldsymbol{\gamma}) \| \xrightarrow{p} 0$ for a continuous $\mathbf{A}(\cdot, \cdot)$ with $\mathbf{A} = \mathbf{A}(\boldsymbol{\beta}_0, \boldsymbol{\gamma}^*)$ non-singular;
(E4) $n^{-1/2} [\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}) - \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*)] \xrightarrow{p} \mathbf{0}$.
Then:
(i) for every sufficiently small $r > 0$, with probability tending to one, $\mathbf{U}_n(\boldsymbol{\beta}; \hat{\boldsymbol{\gamma}}) = \mathbf{0}$ has exactly one root $\hat{\boldsymbol{\beta}}$ in the ball $\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le r$, and $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$;
(ii) $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1\prime})$;
(iii) if moreover $n^{-1} \sum_i \boldsymbol{\psi}_i \boldsymbol{\psi}_i'$ converges in probability, uniformly on $\mathcal{N} \times \mathcal{M}$, to a continuous limit equal to $\mathbf{B}$ at $(\boldsymbol{\beta}_0, \boldsymbol{\gamma}^*)$, then with $\hat{\mathbf{A}} = n^{-1} \sum_i \mathbf{H}_i(\hat{\boldsymbol{\beta}}; \hat{\boldsymbol{\gamma}})$ and $\hat{\mathbf{B}} = n^{-1} \sum_i \boldsymbol{\psi}_i \boldsymbol{\psi}_i'(\hat{\boldsymbol{\beta}}; \hat{\boldsymbol{\gamma}})$, the sandwich $\hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-1\prime} \xrightarrow{p} \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1\prime}$.

Proof. [Supplement]

Step 1. Set-up. Let $c = \|\mathbf{A}^{-1}\|$. Choose $r > 0$ and a neighbourhood $\mathcal{M}' \subseteq \mathcal{M}$ of $\boldsymbol{\gamma}^*$ with $\bar{B}_r = \{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le r\} \subseteq \mathcal{N}$ and $\|\mathbf{A}(\boldsymbol{\beta}, \boldsymbol{\gamma}) - \mathbf{A}\| \le 1/(4c)$ on $\bar{B}_r \times \mathcal{M}'$. Let ε_n be the supremum in (E3) and

$$E_n = \left\{ \hat{\boldsymbol{\gamma}} \in \mathcal{M}', \quad \varepsilon_n \le \tfrac{1}{4c}, \quad \|n^{-1} \mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}})\| \le \tfrac{r}{2c} \right\}.$$

By (E1)–(E2) and Chebyshev, $n^{-1} \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) \xrightarrow{p} \mathbf{0}$, hence by (E4) also with $\hat{\boldsymbol{\gamma}}$; so $P(E_n) \to 1$.

Step 2. Linearization. For $\boldsymbol{\beta} \in \bar{B}_r$ (convex),

$$n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}}) = n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}}) - \mathbf{A}_n(\boldsymbol{\beta})(\boldsymbol{\beta}-\boldsymbol{\beta}_0), \qquad \mathbf{A}_n(\boldsymbol{\beta}) = \int_0^1 n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta}_0 + s(\boldsymbol{\beta}-\boldsymbol{\beta}_0);\hat{\boldsymbol{\gamma}})\, ds,$$

and on E_n, $\|\mathbf{A}_n(\boldsymbol{\beta}) - \mathbf{A}\| \le \varepsilon_n + \frac{1}{4c} \le \frac{1}{2c}$ on $\bar{B}_r$.

Step 3. Existence. The fixed points of $F(\boldsymbol{\beta}) = \boldsymbol{\beta} + \mathbf{A}^{-1}n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})$ are roots, and by Step 2

$$F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0 = \mathbf{A}^{-1}n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}}) + \mathbf{A}^{-1}(\mathbf{A} - \mathbf{A}_n(\boldsymbol{\beta}))(\boldsymbol{\beta}-\boldsymbol{\beta}_0),$$

so on E_n, $\|F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0\| \le c \cdot \frac{r}{2c} + c \cdot \frac{1}{2c} \cdot r = r$. By Brouwer's theorem F has a fixed point $\hat{\boldsymbol{\beta}} \in \bar{B}_r$.

Step 4. Uniqueness. Two roots $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2 \in \bar{B}_r$ give $\mathbf{0} = \tilde{\mathbf{A}}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_2)$ with $\|\tilde{\mathbf{A}} - \mathbf{A}\| \le \frac{1}{2c}$; since $\|\mathbf{A}^{-1}(\tilde{\mathbf{A}} - \mathbf{A})\| \le \frac{1}{2}$, $\tilde{\mathbf{A}}$ is invertible (Neumann series) and $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$.

Step 5. Consistency. The argument holds for every $r' \le r$, and by uniqueness the roots coincide, so $P(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\| \le r') \to 1$ for all $r' > 0$. This proves (i).

Step 6. Asymptotic normality. On E_n, $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \mathbf{A}_n(\hat{\boldsymbol{\beta}})^{-1}\, n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})$. By (E3), continuity and consistency, $\mathbf{A}_n(\hat{\boldsymbol{\beta}}) \xrightarrow{p} \mathbf{A}$. By (E4), $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}}) = n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*) + o_p(1)$, a normalized sum of independent mean-zero vectors; for fixed $\boldsymbol{\lambda} \neq \mathbf{0}$ Lyapunov's condition

$$\frac{\sum_i E|\boldsymbol{\lambda}'\boldsymbol{\psi}_i|^{2+\delta}}{\left(\sum_i \mathrm{Var}(\boldsymbol{\lambda}'\boldsymbol{\psi}_i)\right)^{1+\delta/2}} = \frac{O(n)}{\left(n\,\boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda}\,(1+o(1))\right)^{1+\delta/2}} \to 0$$

holds, so by Cramér–Wold $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*) \xrightarrow{d} N(\mathbf{0},\mathbf{B})$, and Slutsky gives (ii).

Step 7. Sandwich. $\hat{\mathbf{A}} \xrightarrow{p} \mathbf{A}$ as in Step 6 and $\hat{\mathbf{B}} \xrightarrow{p} \mathbf{B}$ by the extra assumption; continuity of inversion and multiplication proves (iii). ■

Theorem 11.9 (Consistency and asymptotic normality of QL estimators)

Suppose the mean model $g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$ is correct at $\boldsymbol{\beta}_0$ (the variance function v may be wrong) and that conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i = (\partial\mu_i/\partial\boldsymbol{\beta})(y_i - \mu_i)/v(\mu_i)$ (they hold, e.g., for bounded covariates, smooth g and v, $v > 0$, and bounded fourth moments of y_i). Then the QL estimator is consistent and asymptotically normal, with

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}\Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\frac{\mathrm{Var}(Y_i)}{v(\mu_i)^2}\frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]\mathbf{V}, \qquad \mathbf{V} = \Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\frac{1}{v(\mu_i)}\frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]^{-1}. \tag{11.8}$$

If $\mathrm{Var}(Y_i) = v(\mu_i)$, this reduces to $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}$, the GLM formula.

Proof.

Step 1. (E1) holds since $E(y_i) = \mu_i(\boldsymbol{\beta}_0)$ and the coefficient of $y_i - \mu_i$ is nonrandom; (E4) is vacuous.

Step 2. $\mathbf{H}_i = v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}') - (\partial[v(\mu_i)^{-1}\partial\mu_i/\partial\boldsymbol{\beta}]/\partial\boldsymbol{\beta}')(y_i - \mu_i)$; the second term has mean zero at $\boldsymbol{\beta}_0$, giving $\mathbf{A} = \lim n^{-1}\sum_i v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$, and $\mathrm{Cov}(\boldsymbol{\psi}_i) = v(\mu_i)^{-2}\,\mathrm{Var}(Y_i)(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$ gives $\mathbf{B}$.

Step 3. Lemma 11.8(ii) gives $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx n^{-1}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, which is (11.8); if $\mathrm{Var}(Y_i) = v(\mu_i)$ then $\mathbf{B} = \mathbf{A}$ and it collapses to $\mathbf{V}$. ■

11.4.2 GEE methodology: technical details

For cluster i let $\mathbf{y}_i = (y_{i1},\dots,y_{iT_i})'$, $\boldsymbol{\mu}_i = E(\mathbf{Y}_i)$, $\mathbf{X}_i$ the $T_i \times p$ matrix with rows $\mathbf{x}_{it}'$, and $\eta_{it} = g(\mu_{it}) = \mathbf{x}_{it}'\boldsymbol{\beta}$. Each margin has

$$f(y_{it};\theta_{it},\phi) = \exp\{[y_{it}\theta_{it} - b(\theta_{it})]/\phi + c(y_{it},\phi)\},$$

so $\mu_{it} = b'(\theta_{it})$ and $\operatorname{Var}(Y_{it}) = \phi\, b''(\theta_{it})$.

Definition (Generalized estimating equations)

Let $\mathbf{B}_i = \operatorname{diag}\big(b''(\theta_{i1}), \dots, b''(\theta_{iT_i})\big)$ and let $\mathbf{R}(\boldsymbol{\alpha})$ be a working correlation matrix. The **working covariance** is

$$\mathbf{V}_i = \phi\, \mathbf{B}_i^{1/2} \mathbf{R}(\boldsymbol{\alpha}) \mathbf{B}_i^{1/2} \tag{11.9}$$

($= \operatorname{Cov}(\mathbf{Y}_i)$ if $\mathbf{R}$ is the true correlation). Let $\boldsymbol{\Delta}_i = \operatorname{diag}(\partial\theta_{it}/\partial\eta_{it})$ (the identity for the canonical link) and $\mathbf{D}_i = \partial\boldsymbol{\mu}_i/\partial\boldsymbol{\beta}' = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ ($T_i \times p$). The GEE estimator $\hat{\boldsymbol{\beta}}$ solves

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}. \tag{11.10}$$

$\mathbf{D}_i = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ is the chain rule, and the scalar ϕ cancels from (11.10).

Proposition 11.10 (Independence working correlation gives the GLM fit)

With $\mathbf{R} = \mathbf{I}$, (11.10) becomes $\sum_i \mathbf{X}_i'\boldsymbol{\Delta}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ (for the canonical link, $\sum_i \mathbf{X}_i'(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$): the ordinary GLM likelihood equations treating all y_{it} as independent.

Proof. With $\mathbf{V}_i = \phi\mathbf{B}_i$, $\mathbf{D}_i'\mathbf{V}_i^{-1} = \mathbf{X}_i'\boldsymbol{\Delta}_i\mathbf{B}_i(\phi\mathbf{B}_i)^{-1} = \phi^{-1}\mathbf{X}_i'\boldsymbol{\Delta}_i$; drop $1/\phi$ and compare with Proposition 11.7(a) summed over all (i,t). ■

Definition (Moment estimators of the working parameters)

With Pearson residuals $e_{it} = (y_{it} - \hat{\mu}_{it})/b''(\hat{\theta}_{it})^{1/2}$, $N = \sum_i T_i$ (Liang and Zeger 1986):

$$\hat{\phi} = \frac{1}{N-p} \sum_i \sum_t e_{it}^2, \qquad \hat{\alpha}_{\text{exch}} = \frac{\sum_i \sum_{t<s} e_{it}e_{is}}{\hat{\phi}\big[\sum_i T_i(T_i-1)/2 - p\big]},$$

$$\hat{\alpha}_{\text{AR1}} = \frac{\sum_i \sum_{t \le T_i - 1} e_{it}e_{i,t+1}}{\hat{\phi}\big[\sum_i (T_i - 1) - p\big]}, \qquad \hat{R}_{ts} = \frac{1}{n\hat{\phi}} \sum_i e_{it}e_{is} \ \text{ (unstructured)}.$$

The algorithm alternates a modified Fisher scoring step for $\boldsymbol{\beta}$ given $(\boldsymbol{\alpha}, \phi)$,

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} + \Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1} \sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i),$$

with moment updates of $(\boldsymbol{\alpha}, \phi)$ from the current residuals.

Theorem 11.11 (Liang–Zeger: consistency and asymptotic normality of GEE)

Assume:
(G1) the marginal mean model is correct: $E(\mathbf{Y}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}_0)$ for all i (the working correlation may be wrong);
(G2) $\sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = O_p(1)$ for some $\boldsymbol{\alpha}^*$ (not necessarily a "true" value);
(G3) conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i(\boldsymbol{\beta}; \boldsymbol{\alpha}) = \mathbf{D}_i'\mathbf{V}_i^{-1}(\boldsymbol{\alpha})(\mathbf{y}_i - \boldsymbol{\mu}_i)$, and $n^{-1}\sum_i \partial\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha})/\partial\boldsymbol{\alpha}' \xrightarrow{p} \mathbf{0}$ uniformly for $\boldsymbol{\alpha}$ near $\boldsymbol{\alpha}^*$.
Then, as the number of clusters $n \to \infty$, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_G)$, where $\mathbf{V}_G = \lim_n \mathbf{V}_{G,n}$,

$$\mathbf{V}_{G,n} = n\Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1}\Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\operatorname{Cov}(\mathbf{Y}_i)\mathbf{V}_i^{-1}\mathbf{D}_i\Big]\Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1},$$

evaluated at $(\boldsymbol{\beta}_0, \boldsymbol{\alpha}^*)$. The sandwich estimator $\hat{\mathbf{V}}_{G,n}/n$, which substitutes $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\alpha}}$, $\hat{\phi}$ and replaces $\mathrm{Cov}(\mathbf{Y}_i)$ by $(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)'$, is consistent for $\mathrm{Cov}(\hat{\boldsymbol{\beta}})$.

Proof. [Supplement] We verify the hypotheses of Lemma 11.8 with nuisance $\boldsymbol{\gamma} = \boldsymbol{\alpha}$ (ϕ cancels from (11.10)).

Step 1. (E1). At $\boldsymbol{\beta}_0$, $\mathbf{D}_i$ and $\mathbf{V}_i(\boldsymbol{\alpha}^*)$ are nonrandom and $E(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ by (G1), so $E\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*) = \mathbf{0}$ *for any* $\boldsymbol{\alpha}^*$: a wrong working correlation does not destroy consistency.

Step 2. $\mathbf{A}$ and $\mathbf{B}$. Differentiating,

$$\mathbf{H}_i = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i - \sum_j \Big[\frac{\partial (\mathbf{D}_i' \mathbf{V}_i^{-1})}{\partial \beta_j} (\mathbf{y}_i - \boldsymbol{\mu}_i) \Big] \mathbf{e}_j',$$

whose second term has mean zero at $\boldsymbol{\beta}_0$, so $\mathbf{A} = \lim n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$; and $\mathrm{Cov}(\boldsymbol{\psi}_i) = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathrm{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i$ averages to $\mathbf{B}$. (E2), (E3) hold by (G3).

Step 3. (E4). By the mean value theorem, for some $\bar{\boldsymbol{\alpha}}$ between $\hat{\boldsymbol{\alpha}}$ and $\boldsymbol{\alpha}^*$ (componentwise),

$$n^{-1/2} [\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\alpha}}) - \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*)] = \Big[n^{-1} \sum_i \frac{\partial \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \bar{\boldsymbol{\alpha}})}{\partial \boldsymbol{\alpha}'} \Big] \sqrt{n} (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*).$$

Each $\partial \boldsymbol{\psi}_i / \partial \alpha_k = \mathbf{D}_i' (\partial \mathbf{V}_i^{-1} / \partial \alpha_k)(\mathbf{y}_i - \boldsymbol{\mu}_i)$ has mean zero at $\boldsymbol{\beta}_0$; by (G3) the bracket is $o_p(1)$ and by (G2) the last factor is $O_p(1)$, so the product is $o_p(1)$: estimating $\boldsymbol{\alpha}$ has no first-order effect.

Step 4. Lemma 11.8(i)–(ii) gives consistency and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$ with $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \lim \mathbf{V}_{G,n}$. Lemma 11.8(iii) gives consistency of the sandwich with $\hat{\mathbf{B}} = n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)' \mathbf{V}_i^{-1} \mathbf{D}_i$ at $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}})$; replacing $\hat{\mathbf{A}}$ by $n^{-1} \sum \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ at the estimates changes it by $o_p(1)$ (Step 2). ■

Corollary 11.12 (Model-based covariance)

If the working covariance is correct, $\mathrm{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, then $\mathbf{V}_{G,n}/n = \big(\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \big)^{-1}$.

Proof. Substituting $\mathrm{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, the middle factor equals $\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ and cancels one outer inverse. ■

Theorem 11.13 (Optimality of the true covariance as weight)

Consider all estimating equations $\sum_i \mathbf{D}_i' \mathbf{W}_i (\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ with nonrandom positive definite weights $\mathbf{W}_i$, and let $\boldsymbol{\Sigma}_i = \mathrm{Cov}(\mathbf{Y}_i)$. Their asymptotic covariance satisfies, in the Loewner order,

$$\mathbf{G}^{-1} \mathbf{M} \mathbf{G}^{-1} \;\geq\; \mathbf{F}^{-1}, \qquad \mathbf{G} = \sum_i \mathbf{D}_i' \mathbf{W}_i \mathbf{D}_i, \quad \mathbf{M} = \sum_i \mathbf{D}_i' \mathbf{W}_i \boldsymbol{\Sigma}_i \mathbf{W}_i \mathbf{D}_i, \quad \mathbf{F} = \sum_i \mathbf{D}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i,$$

with equality when $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$. So a correct working covariance gives the most efficient GEE estimator.

Proof. [Supplement]

Step 1. With $\mathbf{K}_i = \begin{pmatrix} \mathbf{W}_i \mathbf{D}_i & \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i \end{pmatrix}$, the matrix $\mathbf{K}_i' \boldsymbol{\Sigma}_i \mathbf{K}_i = \begin{pmatrix} \mathbf{D}_i' \mathbf{W}_i \boldsymbol{\Sigma}_i \mathbf{W}_i \mathbf{D}_i & \mathbf{D}_i' \mathbf{W}_i \mathbf{D}_i \\ \mathbf{D}_i' \mathbf{W}_i \mathbf{D}_i & \mathbf{D}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i \end{pmatrix}$ is positive semidefinite; summing, $\left(\begin{smallmatrix} \mathbf{M} & \mathbf{G} \\ \mathbf{G} & \mathbf{F} \end{smallmatrix}\right) \geq 0$.

Step 2. Since $\mathbf{F} > 0$, the Schur complement $\mathbf{M} - \mathbf{G} \mathbf{F}^{-1} \mathbf{G} \geq 0$.

Step 3. Congruence by the symmetric $\mathbf{G}^{-1}$ gives $\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \geq \mathbf{F}^{-1}$; if $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$ then $\mathbf{G} = \mathbf{M} = \mathbf{F}$. ■

Correlation is an awkward association measure for binary data

The Fréchet bounds $\max(0, p_t + p_s - 1) \leq P(Y_t = Y_s = 1) \leq \min(p_t, p_s)$ restrict $\mathrm{Corr}(Y_t, Y_s)$ by the means (at most 1/3 for $p_t = 0.1$, $p_s = 0.5$). Odds ratios are variation-independent of the means; *alternating logistic regressions* (Carey, Zeger and Diggle 1993) alternate GEE steps for $\boldsymbol{\beta}$ with logistic steps for the log odds ratios.

11.5 Markov chains: transitional modeling

Motivation. Modelling each response given earlier ones factors the joint distribution into transitions, fitted by ordinary multinomial or logistic models.

11.5.1 Transitional models

For a discrete-time, discrete-state process $\{Y_t\}$, **transitional models** describe Y_t given $y_0, \ldots, y_{t-1}$ (and covariates), via

$$f(y_0, \ldots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_0, y_1) \cdots f(y_T \mid y_0, \ldots, y_{T-1}).$$

11.5.2 First-order Markov chains

Definition (Markov chain)

$\{Y_t\}$ is a (first-order) **Markov chain** if Y_{t+1} given $(Y_0, \ldots, Y_t)$ depends only on Y_t, and **kth-order** if only on $(Y_{t-k+1}, \ldots, Y_t)$. The **transition probabilities** $\pi_{j|i}(t) = P(Y_t = j \mid Y_{t-1} = i)$, $\sum_j \pi_{j|i}(t) = 1$, form the $I \times I$ transition matrix.

Theorem 11.14 (Markov factorization and its loglinear form)

(a) $\{Y_0, \ldots, Y_T\}$ is a Markov chain if and only if

$$f(y_0, \ldots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_1) \cdots f(y_T \mid y_{T-1}). \tag{11.11}$$

(b) A positive joint distribution is that of a Markov chain if and only if it satisfies the loglinear model $(Y_0Y_1, Y_1Y_2, \ldots, Y_{T-1}Y_T)$, i.e. $\log \pi(y_0, \ldots, y_T) = \sum_{t=1}^{T} h_t(y_{t-1}, y_t)$ for some functions h_t.

Proof. [Supplement]

Step 1. (a) The Markov property turns the general factorization into (11.11). Conversely, summing (11.11) over $y_{t+1}, \ldots, y_T$ gives $f(y_0, \ldots, y_t) = f(y_0) \prod_{s \leq t} f(y_s \mid y_{s-1})$, and the ratio of these for $t+1$ and t is $f(y_{t+1} \mid y_t)$.

Step 2. (b, $\Rightarrow$) Take logs of (11.11), absorbing $\log f(y_0)$ into h_1.

Step 3. (b, $\Leftarrow$) If $\pi(\mathbf{y}) = \prod_t g_t(y_{t-1}, y_t)$, let $G_t(y_t) = \sum_{y_{t+1}, \ldots, y_T} \prod_{s>t} g_s(y_{s-1}, y_s)$ ($G_T \equiv 1$). Then

$$P(y_{t+1} \mid y_0, \ldots, y_t) = \frac{g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{G_t(y_t)},$$

since the factors $\prod_{s\le t} g_s$ cancel; this depends only on (y_t, y_{t+1}). ■

Adding three-factor terms $(Y_{t-2}Y_{t-1}Y_t)$ gives a second-order chain.

Theorem 11.15 (ML estimation of transition probabilities)

Let $n_{ij}(t)$ count transitions $i \to j$ from time $t-1$ to t, $n_{i+}(t) = \sum_j n_{ij}(t)$, and $\{n_{i0}\}$ the initial counts, multinomial with $\{\pi_{i0}\}$. For n independent subjects the likelihood is proportional to

$$\Big[\prod_{i=1}^{I} \pi_{i0}^{n_{i0}}\Big] \prod_{t=1}^{T}\prod_{i=1}^{I}\prod_{j=1}^{I} \pi_{j|i}(t)^{n_{ij}(t)}, \tag{11.12}$$

and the ML estimates are $\hat\pi_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$ (Anderson and Goodman 1957). For a time-homogeneous chain, $\pi_{j|i}(t) \equiv \pi_{j|i}$, the ML estimate is $\hat\pi_{j|i} = \sum_t n_{ij}(t) / \sum_t n_{i+}(t)$.

Proof.

Step 1. Each subject contributes $\pi_{y_0 0}\prod_t \pi_{y_t|y_{t-1}}(t)$ by (11.11); grouping factors gives (11.12).

Step 2. [Supplement] The likelihood factors into separate multinomial kernels $\prod_j \pi_{j|i}(t)^{n_{ij}(t)}$ with unrelated parameters, each maximized separately.

Step 3. For a kernel with counts n_j, $m = \sum_j n_j > 0$, Jensen's inequality gives

$$\sum_j n_j \log \pi_j - \sum_j n_j \log \frac{n_j}{m} = m \sum_j \frac{n_j}{m} \log \frac{\pi_j}{n_j/m} \le m \log \sum_j \pi_j = 0,$$

with equality iff $\pi_j = n_j/m$.

Step 4. Under time homogeneity the kernels for fixed i merge, with counts $\sum_t n_{ij}(t)$. ■

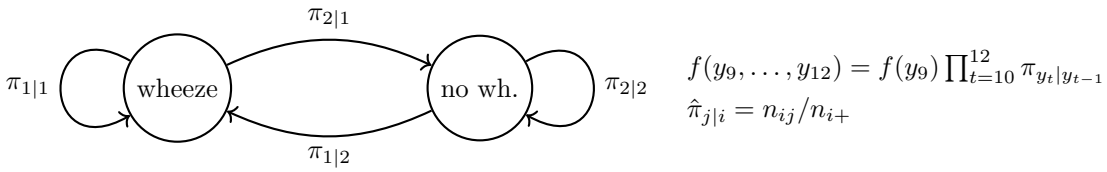


Figure 11.2. A two-state first-order Markov chain.

11.5.3 Respiratory illness example

Example 11.16 (Wheeze at ages 9–12)

Harvard air-pollution study: wheeze at ages 9–12 (2^4 table, $n = 1019$). First-order ($G^2 = 122.9$, df $= 8$) and second-order chains ($G^2 = 23.9$, df $= 4$) fit poorly, perhaps through subject heterogeneity. All pairwise associations fit ($G^2 = 1.5$, df $= 5$), as does one log odds ratio for pairs one year apart (1.75) and another for pairs further apart (1.04) ($G^2 = 2.3$, df $= 9$).

11.5.4 Transitional models with explanatory variables

With covariates, $f(y_1,\ldots,y_T;\mathbf{x}) = f(y_1;\mathbf{x})\prod_{t\ge 2} f(y_t \mid y_1,\ldots,y_{t-1};\mathbf{x})$, and for binary y the **regressive logistic model** (Bonney 1987) is

$$f(y_t \mid y_1,\ldots,y_{t-1};\mathbf{x}_t) = \frac{\exp\left[y_t(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1}y_{t-1} + \boldsymbol{\beta}'\mathbf{x}_t)\right]}{1+\exp(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1}y_{t-1} + \boldsymbol{\beta}'\mathbf{x}_t)}, \qquad y_t = 0, 1.$$

First-order Markov structure drops $y_1, \ldots, y_{t-2}$. Unlike marginal effects, $\boldsymbol{\beta}$ depends on which past responses are conditioned on.

Proposition 11.17 (Fitting transitional models by ordinary logistic regression)

Given the first response(s), the likelihood is a product of Bernoulli likelihoods with the stated logits over subjects and transitions, so conditional ML is ordinary logistic regression with each transition an independent observation and past responses as predictors.

Proof. [Supplement] Subject i contributes $f(y_{i1}; \mathbf{x}) \prod_{t \geq 2} f(y_{it} \mid y_{i1}, \ldots, y_{i,t-1}; \mathbf{x}_{it})$. Conditioning on the first factor, each remaining factor is $\text{expit}(\eta_{it})^{y_{it}}[1 - \text{expit}(\eta_{it})]^{1-y_{it}}$ with η_{it} linear in the parameters and known regressors (past responses, $\mathbf{x}_{it}$). Independence across subjects makes the product over all (i, t) a logistic-regression likelihood with one observation per transition. ■

11.5.5 Child's respiratory illness and maternal smoking

Example 11.18 (Respiratory illness and maternal smoking)

Respiratory illness at ages 7–10; $s = 1$ if the mother smoked. With three transitions per child (12 binomials over (s, t, y_{t-1})), the ML fit is

$$\text{logit}\, \hat{P}(Y_t = 1) = -0.293 + 0.296s - 0.243t + 2.211y_{t-1}, \qquad t = 8, 9, 10,$$

with SEs $(0.846, 0.156, 0.095, 0.158)$ and $G^2 = 3.1$ (df $= 8$). The previous response has a strong effect; the smoking effect is marginal (LR 3.55, $P = 0.06$).